Smart Predictive Maintenance for Aircraft Engines Using IoT and Machine Learning

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Abstract

Predictive maintenance is crucial in the aviation industry to prevent unexpected failures and ensure operational efficiency. The study on Smart Predictive Maintenance for Aircraft Engines using IoT and Machine Learning, utilizes sensor data from aircraft engines to predict potential failures before they occur. The dataset undergoes preprocessing, including duplicate removal, feature engineering, and data normalization. Various machine learning models, such as Gradient Boosting, Random Forest, and AdaBoost, are evaluated to determine the most effective failure prediction system. Hyperparameter tuning using Optuna enhances model performance, and a Streamlit-based interface provides real-time analysis and maintenance scheduling. The proposed system helps optimize aircraft maintenance strategies, reducing costs and enhancing safety.

Keywords: Predictive Maintenance, Aircraft Engines, Machine Learning, IoT

1 Introduction

The aviation industry relies on highly efficient and reliable aircraft engines to ensure safety, minimize operational costs, and prevent unexpected failures. Traditional maintenance strategies, such as reactive maintenance (fixing after failure) and scheduled maintenance (fixed time intervals), often lead to unnecessary maintenance or

unexpected breakdowns. These methods lack adaptability and real-time monitoring, resulting in increased costs and downtime. To address this, predictive maintenance leverages IoT sensors and machine learning to analyze engine performance in real-time, detect early signs of failure, and optimize maintenance schedules pro-actively.

With advancements in machine learning (ML) and IoT, predictive maintenance has become a game-changer in the aviation industry. By continuously monitoring engine health metrics, ML algorithms can identify patterns in sensor data, allowing early detection of component failures. The ability to forecast potential issues reduces unplanned maintenance costs, enhances engine lifespan, and ensures operational efficiency. Moreover, hyperparameter tuning techniques, such as Optuna, improve the performance of predictive models, making failure detection more accurate and reliable.

This study focuses on implementing a real-time failure prediction system using machine learning models. It processes sensor-based aircraft engine data, applies feature engineering and hyper-parameter tuning, and evaluates multiple classifiers, including Gradient Boosting, Random Forest, and AdaBoost. A Streamlit-based interactive interface enables aviation engineers to monitor engine health, visualize failure probabilities, and schedule maintenance efficiently. This intelligent system ensures cost-effective, data-driven decision-making, ultimately improving aircraft safety and performance.

2 Related Works

Predictive maintenance has gained significant attention in recent years, leveraging machine learning and IoT technologies to enhance failure prediction and optimize maintenance strategies. This literature survey presents a summary of ten relevant studies that contribute to the advancement of predictive maintenance, particularly in the aviation industry.

Lazzaro et al. [1] conducted a detailed study on predictive maintenance algorithms in smart manufacturing, focusing on chip form classification using edge machine learning. The research highlights the importance of real-time predictive analytics in industrial applications, demonstrating that machine learning models deployed at the edge can efficiently detect faults and optimize maintenance schedules. The study underscores the potential of AI-driven fault detection, which is applicable to aircraft engine maintenance by processing sensor data in real-time.

Chen et al. [2] proposed an AI-based human-centric decision support framework for predictive maintenance in asset management. The study emphasizes the role of artificial intelligence in improving maintenance strategies, especially under challenging conditions such as the COVID-19 pandemic. The framework integrates human expertise with AI-driven insights, ensuring efficient decision-making for maintenance planning. This approach is relevant to aircraft engines, where maintenance decisions require a balance between automated predictions and expert evaluations.

Proto et al. [3] introduced REDTag, a predictive maintenance framework designed for parcel delivery services. The study explores the use of machine learning models to predict component failures and optimize maintenance schedules in logistics operations. While the primary focus is on fleet management, the methodology can be extended

to aircraft engine maintenance by applying similar sensor-based failure prediction techniques.

In [4], Feng and Li investigated reinforcement learning (RL) techniques for predictive maintenance decision-making in multistage production systems. The study demonstrates how RL algorithms can learn optimal maintenance policies over time, improving efficiency and reducing operational costs. In the context of aircraft engines, RL-based models could dynamically adjust maintenance schedules based on real-time engine performance data, ensuring better resource utilization.

Mi et al. [5] developed an intelligent scheduling system for predictive maintenance in complex industrial equipment. Their approach integrates IoT-based sensor monitoring and AI-driven scheduling algorithms to optimize maintenance workflows. The study highlights the importance of real-time data processing, a key aspect this work where aircraft engine sensors continuously monitor engine health, enabling proactive failure prevention.

A systematic review of machine learning models used for predictive maintenance in aircraft engines is presented in [6]. The research categorizes different ML techniques based on interpretability and accuracy, highlighting the strengths and weaknesses of Random Forest, Gradient Boosting, and Neural Networks. The findings align with this work, where multiple ML models are tested and optimized using Optuna to achieve high failure prediction accuracy.

In [7], authors explored the trustworthiness and key components of AI-driven predictive maintenance systems. Their study discusses the challenges in AI adoption, including model transparency, explainability, and reliability. These insights are crucial for the aviation industry, where regulatory compliance and trust in AI-driven maintenance decisions play a vital role in adoption.

The work in [8] examined the role of machine learning in sustainable smart manufacturing under Industry 4.0. The study emphasizes the importance of predictive analytics in reducing unplanned downtimes and improving operational efficiency. The methodologies used in this research can be extended to aircraft predictive maintenance, where minimizing downtime and optimizing maintenance resources is critical.

Forouzanfar et al. [9] proposed model-based control allocation strategies to enhance predictive maintenance of saturated actuators. Their research highlights the integration of control systems with predictive maintenance models, ensuring optimal performance and extending equipment lifespan. In aircraft engines, such strategies can be used to monitor actuator health and trigger maintenance alerts when performance degrades.

Udo and Muhammad [10] applied data-driven predictive maintenance for wind turbines using SCADA data. The study focuses on sensor-based fault detection, showing how large-scale sensor data analysis can help in early failure prediction. This approach is directly relevant to aircraft engines, where real-time sensor monitoring is used to detect engine anomalies and schedule preventive maintenance.

3 Methodology for Predictive Maintenance scheduling

This study follows a structured approach to analyze engine performance data, predict failures, and assist in proactive maintenance planning. The implementation consists of several stages, including data preprocessing, feature engineering, model selection, hyperparameter tuning, and deployment using Streamlit community cloud platform.

3.1 Data Collection and Understanding

It utilizes a dataset comprised of time-series sensor readings collected from aircraft engines. These readings encompass a variety of metrics, such as temperature, pressure, and vibration levels, providing a comprehensive view of engine performance. Crucially, each data instance is also associated with a failure label, which indicates whether or not an engine failure was recorded at that particular time. For efficient data manipulation and analysis, the dataset is loaded into a Pandas DataFrame, enabling subsequent processing and modelling. Fig 1 shows a sample dataset

	A	B	C	D	E	F	G	H	I	J	K	L
1	date	device	failure	metric1	metric2	metric3	metric4	metric5	metric6	metric7	metric8	metric9
2	01/01/2015	S1F0108S	0	215630672	55	0	52	6	407438	0	0	7
3	01/01/2015	S1F0166B	0	61370680	0	3	0	6	403174	0	0	0
4	01/01/2015	S1F01E6Y	0	173295968	0	0	0	12	237394	0	0	0
5	01/01/2015	S1F01JE0	0	79694024	0	0	0	6	410186	0	0	0
6	01/01/2015	S1F01R2B	0	135970480	0	0	0	15	313173	0	0	3
7	01/01/2015	S1F01TD5	0	68837488	0	0	41	6	413535	0	0	1
8	01/01/2015	S1F01XDJ	0	227721632	0	0	0	8	402525	0	0	0
9	01/01/2015	S1F023H2	0	141503600	0	0	1	19	494462	16	16	3
10	01/01/2015	S1F02A0J	0	8217840	0	1	0	14	311869	0	0	0
11	01/01/2015	S1F02DZ2	0	116440096	0	378	9	9	407905	0	0	170
12	01/01/2015	S1F02EVN	0	112348104	0	0	0	7	388146	0	0	1
13	01/01/2015	S1F02L38	0	223938928	0	0	0	2	215169	0	0	8
14	01/01/2015	S1F02MGA	0	44399688	0	266	1	6	399286	0	0	2269
15	01/01/2015	S1F02P76	0	104131304	1536	0	175	11	301679	0	0	0
16	01/01/2015	S1F02VAX	0	61019512	168	2	521	3	380496	0	0	3
17	01/01/2015	S1F02WFT	0	44348552	6150	14	1074	11	249515	0	0	21
18	01/01/2015	S1F0318A	0	35018688	0	0	0	9	394890	0	0	5
19	01/01/2015	S1F0322R	0	34540712	0	0	0	9	411399	0	0	0
20	01/01/2015	S1F0330P	0	125539768	0	0	12	14	297284	0	0	5

Fig. 1 Sample Dataset

3.2 Exploratory Data Analysis (EDA)

Prior to the training of machine learning models, exploratory data analysis (EDA) is conducted to gain a thorough understanding of the dataset's characteristics. This process begins with duplicate removal, ensuring data integrity by eliminating redundant entries. Visualization techniques are employed to analyze feature distributions; histograms and scatter plots reveal the underlying patterns and correlations within the sensor data. Outlier detection is addressed through log transformations, which normalize specific metrics and mitigate the impact of extreme values. Fig 2 depicts the distribution of failure with respect to different device informations available in the

dataset. Fig 3 and 4 show the distribution of different metrics available in the dataset in the case of no failure and failure respectively. Furthermore, correlation analysis is performed, with a heatmap generated to visually represent the relationships between the various engine parameters, allowing for the identification of significant dependencies.

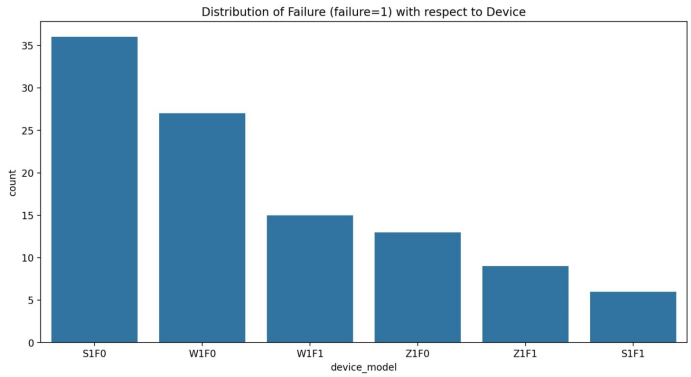

Fig. 2 Distribution on failure w.r.t different devices

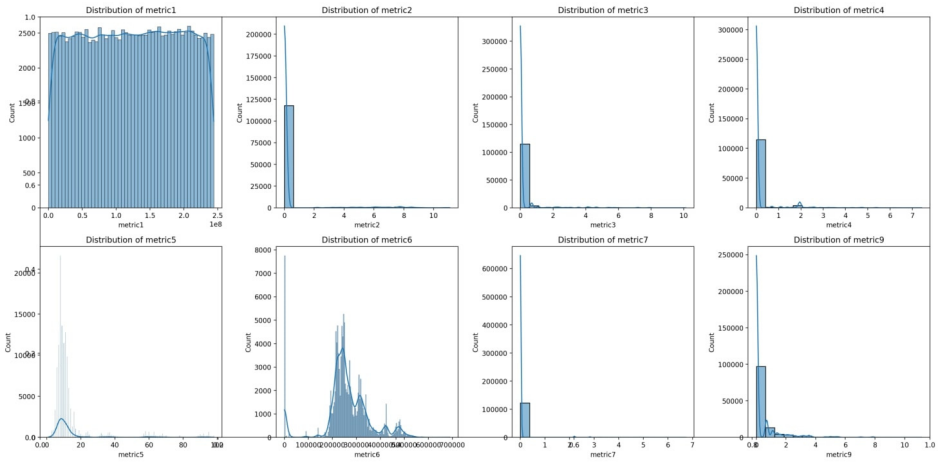

Fig. 3 Distribution of different metrics with failure = 0

3.3 Data Preprocessing

To maintain data integrity and enhance model performance, a series of preprocessing steps are implemented. Missing values, which could skew analysis, are addressed through either imputation or removal, depending on the extent and nature of the missing data. Feature engineering is then performed to extract meaningful insights.

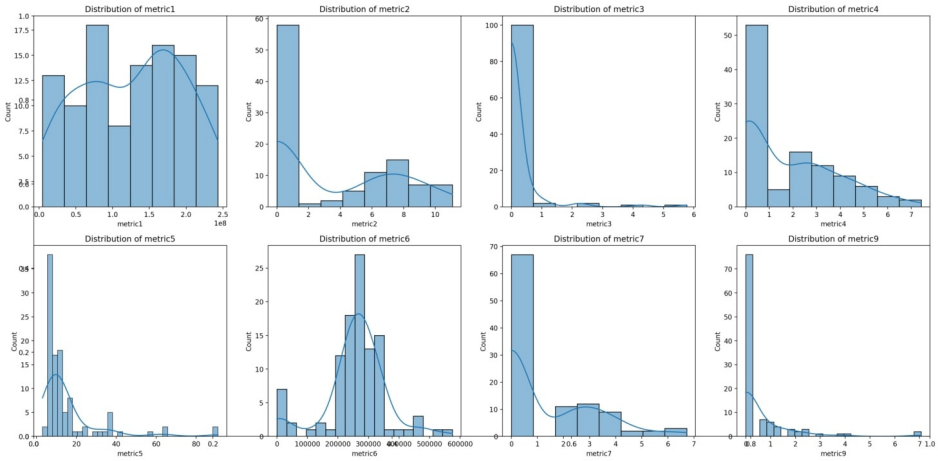

Fig. 4 Distribution of metrics with failure = 1

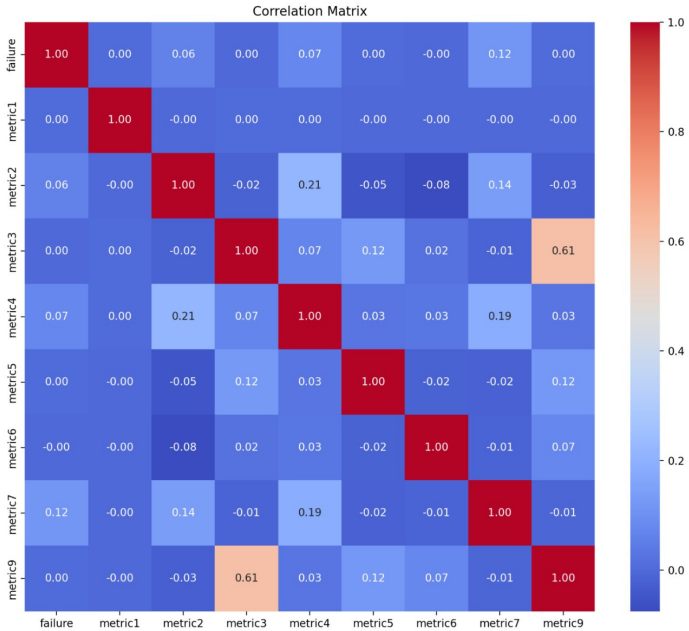

Fig. 5 Correlation Matrix

Specifically, the date column is transformed into month, week, and day_of_week representations, enabling the model to capture temporal patterns that might influence engine failures, see fig 6. Furthermore, the engine ID is dissected into device_model and device_rest components, facilitating a more granular analysis of failure trends across

different engine configurations. Lastly, if any categorical variables are present, one-hot encoding is applied to convert them into numerical representations, making them compatible with machine learning algorithms.

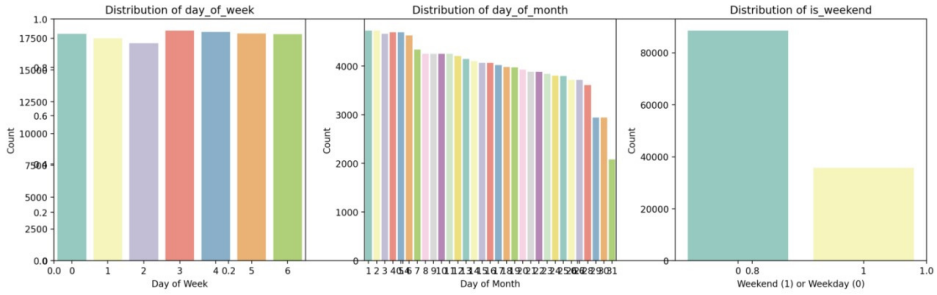

Fig. 6 Distribution of day of week, month and weekend

3.4 Data Balancing Using Undersampling

The dataset exhibits a potential imbalance, characterized by a significantly lower count of failure instances compared to non-failure instances. This imbalance can lead to biased models that predominantly favor the majority class. Fig 7 shows undersampled data. To mitigate this issue, random undersampling (RUS) is employed using the imblearn library. This technique effectively balances the dataset by reducing the number of non-failure instances to match the number of failure instances, ensuring that the model receives a more equitable representation of both classes during training. This balancing act is crucial for preventing the model from developing a bias towards the majority class and improving its ability to accurately predict failures.

3.5 Train-Test Split and Data Normalization

Following data preprocessing and balancing, the dataset is partitioned into distinct training and testing sets to evaluate model performance. Specifically, 80% of the data is allocated for training the models, while the remaining 20% is reserved for testing their predictive capabilities. To ensure that all sensor metrics contribute equitably during model training, standardization is performed using the StandardScaler() from scikit-learn. This process scales the sensor data, transforming it to have a mean of zero and a standard deviation of one, effectively placing all features on a comparable scale. This standardization step is crucial for optimizing the performance of many machine learning algorithms, as it prevents features with larger scales from dominating the learning process.

3.6 Model Selection and Evaluation

A diverse range of supervised machine learning models are deployed and assessed to determine the most effective predictor of aircraft engine failures. These models

	metric1	metric2	metric3	metric4	metric5	metric6	metric7	metric9	month	week	day_of_week	day_of_month	is_weekend
14,696	154,998,752	7.1801	0	2.9444	10	306,996	0	0	1	3	5	17	1
6,861	184,069,720	6.271	0	1.6094	9	387,871	3.4965	1.3863	1	2	2	7	0
109,143	150,125,272	0	0	3.1781	91	251,780	2.1972	0	8	32	0	3	0
116,476	72,548,000	0	0	0	5	223,745	0	0	8	35	6	30	1
67,751	143,676,112	0	0	0	9	221,393	0	0	4	15	2	8	0
124,013	154,144,032	0	0	0	9	352,949	2.8332	0	10	43	5	24	1
118,418	41,239,008	0	0	0	7	274,360	0	0	9	37	4	11	0
62,994	40,512,728	0	0	0	14	284,406	0	0	3	13	6	29	1
8,352	9,012,632	0	0	0	12	292,959	0	0	1	2	3	8	0
57,915	74,655,888	0	0	0	8	190,882	0	1.0986	3	12	5	21	1

Fig. 7 Undersampled Data

include the Gradient Boosting Classifier, Random Forest Classifier, AdaBoost Classifier, Extra Trees Classifier, Decision Tree Classifier, K-Nearest Neighbors (KNN), Naive Bayes (both GaussianNB and BernoulliNB), Support Vector Classifier (SVC), Logistic Regression, and Stochastic Gradient Descent (SGD). Each model undergoes a rigorous evaluation process, beginning with training on the prepared training dataset and subsequently making predictions on the unseen test dataset. To quantify model performance, several key metrics are employed:

- Accuracy: This metric measures the overall proportion of correctly predicted instances out of the total number of instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

- Precision: Precision assesses the proportion of correctly predicted positive instances out of all instances predicted as positive.

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

- Recall: Recall measures the proportion of correctly predicted positive instances out of all actual positive instances.

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

- F1-score: The F1-score is the harmonic mean of precision and recall, providing a balanced measure of a model's performance, especially in imbalanced datasets.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{4}$$

3.7 Hyperparameter Tuning using Optuna

To optimize the performance of the chosen machine learning models, hyperparameter tuning is conducted using the Optuna framework. This process begins with the definition of a search space, where key parameters such as n_estimators, max_depth, learning_rate, and others are specified for tuning. Subsequently, the study.optimize (objective, n_trials=100) function is executed, initiating multiple trials to explore the defined search space and identify the optimal combination of hyperparameters. During these trials, Optuna systematically evaluates different parameter configurations, aiming to minimize or maximize a predefined objective function, such as the F1-score. The best-performing models, characterized by their optimal hyperparameter sets, are then selected for generating the final predictions on the test dataset, ensuring that the models are operating at their peak efficiency. Table 1 shows the selection of the best model.

Table 1 Best model

Model: SGDClassifier
Accuracy: 0.8837
Precision: 0.9932
Recall: 0.9167
F1 Score: 0.8980

3.8 Voting Classifier for Improved Performance

To further enhance prediction accuracy and robustness, a Voting Classifier is employed, strategically combining the predictions of the best-performing individual models. This ensemble approach leverages the strengths of multiple models, mitigating the weaknesses of any single model. The final predictions are generated by a majority voting mechanism, where the class predicted by the majority of the constituent models is selected as the final prediction. This aggregation of predictions from diverse models often leads to improved overall performance and a more reliable prediction outcome.

3.9 Deployment using Streamlit

A user-friendly Streamlit web application is developed to facilitate real-time predictive maintenance analysis of aircraft engines. The application's intuitive user interface provides several key features to empower users with actionable insights. Firstly, a comprehensive dataset overview is presented, including the dataset's shape, a summary of

missing values, and key statistical measures. Secondly, a suite of interactive visualizations, such as scatter plots, histograms, and correlation matrices, is provided to enable in-depth data analysis and pattern recognition. Thirdly, the application displays the performance metrics of the trained machine learning models, including accuracy, precision, recall, and F1-score, allowing users to assess the models' reliability. Fourthly, a failure prediction feature allows users to input specific engine metrics and receive a real-time probability of engine failure. Finally, based on the predicted failure probabilities, the application generates maintenance recommendations, highlighting engines that require immediate maintenance actions.

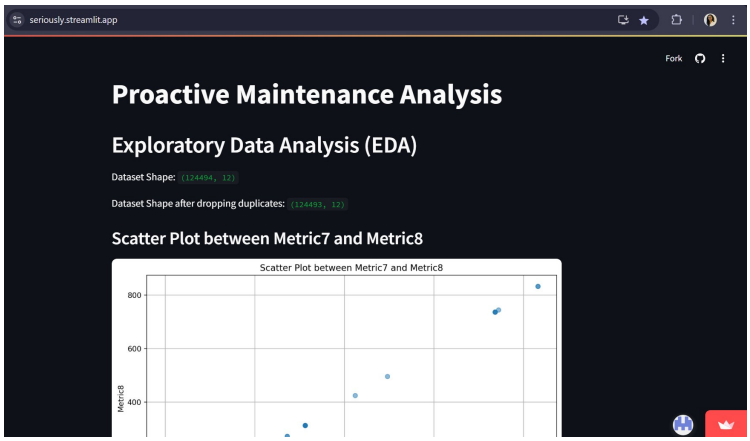

Fig. 8 Screenshot of streamlit application interface.

4 Results and Discussions

The results of this work demonstrate the effectiveness of machine learning-based predictive maintenance in identifying potential aircraft engine failures. By analyzing historical sensor data and applying optimized machine learning models, the system can accurately predict failures before they occur, allowing for proactive maintenance scheduling.

Model Performance and Evaluation:

To determine the best predictive model, multiple classifiers are evaluated using the metrics accuracy, precision, recall, and F1-score. The models performed differently based on how well they identified failure cases, with some excelling in detecting failures (high recall) and others being more precise but missing some failure cases. Among the tested models, Gradient Boosting, Random Forest, and AdaBoost emerged as the top performers, achieving high F1-scores. A voting classifier was also implemented, combining multiple models to improve overall reliability. The confusion matrices helped assess each model?s ability to correctly classify failure and non-failure cases, ensuring that the most effective model was selected for predictive maintenance.

Table 2 depicts the performance evaluation metrics of different machine learning models and figure 9 shows the corresponding confusion matrices.

Table 2 Evaluation metrics for different models

Model	Accuracy	Precision	Recall	F1
SGD	0.8837	0.9932	0.9167	0.8980
Adaboost	0.8605	0.875	0.875	0.875
GaussianNB	0.8605	0.95	0.7917	0.8636
GradientBoost	0.8372	0.9048	0.7917	0.8444
RandomForest	0.8372	0.9048	0.7917	0.8444
ExtraTrees	0.8372	0.84	0.875	0.8751
LogisticRegression	0.8372	0.9474	0.75	0.8372
BernoulliNB	0.814	0.9	0.75	0.8182
SVC	0.814	0.9	0.75	0.8182
DecisionTree	0.7674	0.7917	0.7917	0.7917
KNeighbors	0.7674	0.9375	0.625	0.75

Predictive Maintenance Scheduling and Machines Recommended for Maintenance:

One of the key applications of the system is its ability to recommend maintenance schedules based on real-time failure probability predictions. Using the best-performing machine learning model, the failure probability is predicted for each aircraft engine. A maintenance threshold was set, allowing the system to flag engines with a failure probability above this threshold. Engines identified as high-risk were listed for immediate maintenance to prevent unexpected failures. The system provides a real-time interface where engineers can visualize failure probabilities and make data-driven maintenance decisions.

- The maintenance recommendations help reduce downtime and prevent costly emergency repairs.
- By using sensor-based insights, maintenance teams can focus on at-risk engines rather than scheduling unnecessary maintenance for all engines.
- This approach enhances the efficiency of maintenance operations in aviation.

ROC Curve Analysis:

To further evaluate model performance, the Receiver Operating Characteristic (ROC) Curve is plotted for one of the top-performing models (AdaBoost Classifier). The ROC curve is a crucial tool in classification problems, showing the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR). A model with a high Area Under the Curve (AUC) indicates better performance in distinguishing between failure and non-failure cases. The AUC score for the best models was high, confirming their reliability in making accurate predictions.

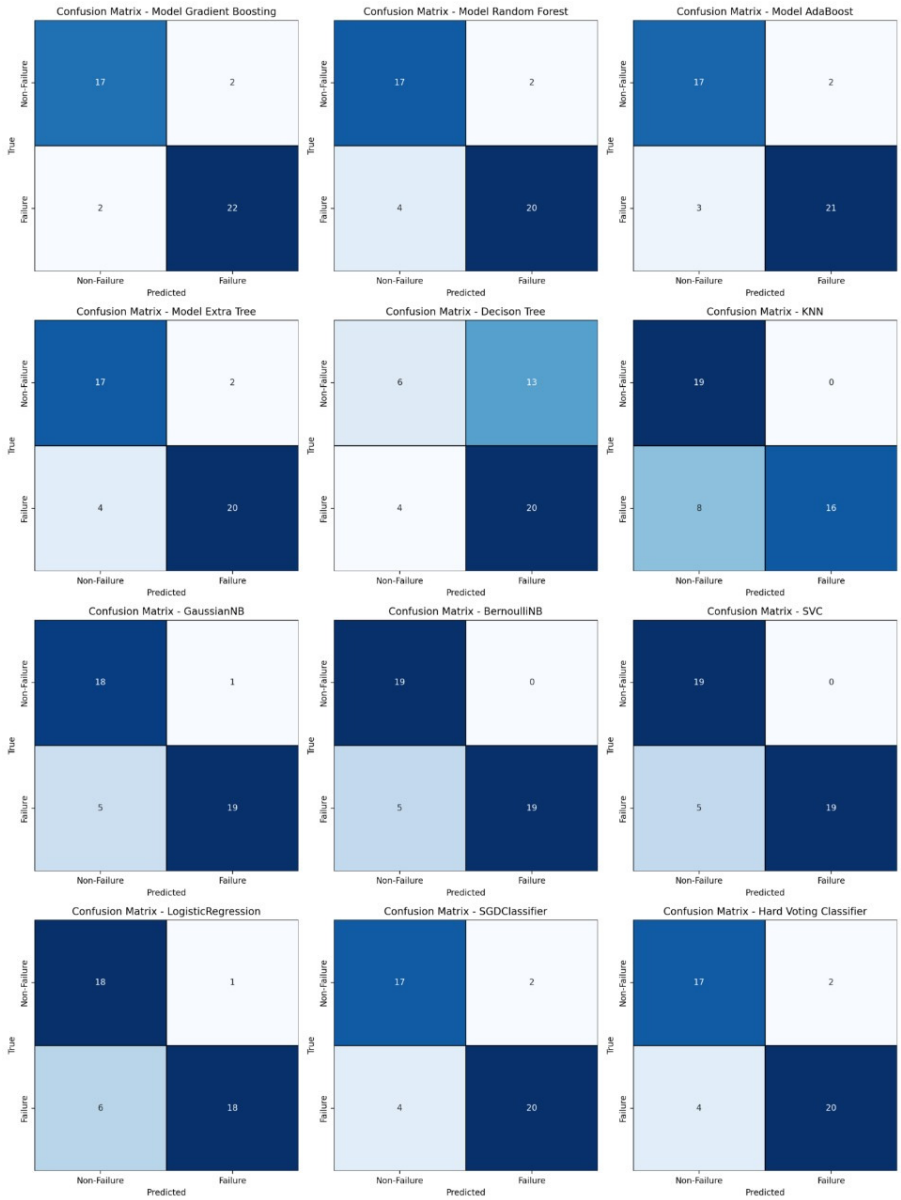

Fig. 9 Confusion matrix of different models.

The ROC curve in figure 10 shows how well the model can differentiate between failing and non-failing engines. A steeper curve with an AUC closer to 1 suggests a strong model with minimal false predictions.

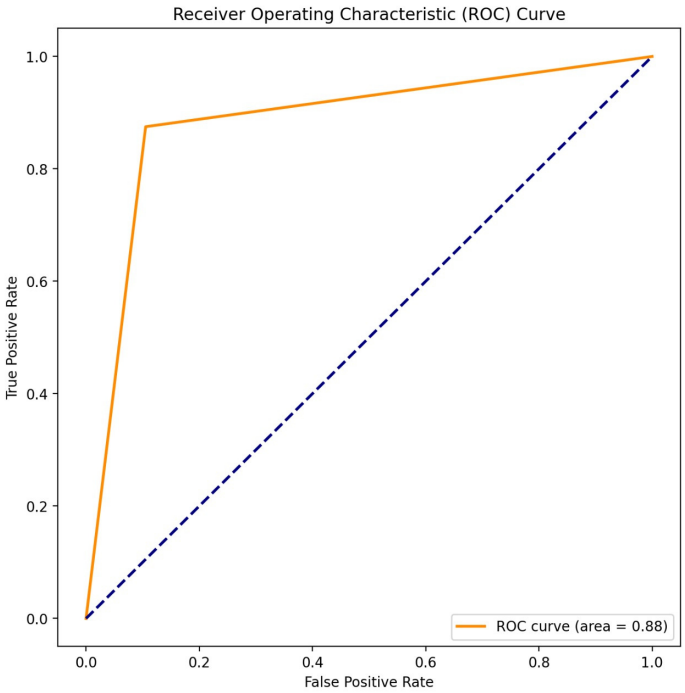

Fig. 10 ROC Curve

5 Conclusion

This work tries to implement a predictive maintenance system for aircraft engines using machine learning and IoT-based sensor data analysis. By leveraging advanced machine learning models and hyperparameter tuning with Optuna, the system accurately predicts engine failures, allowing proactive maintenance scheduling. The integration of a Streamlit-based interface enables real-time monitoring and decision-making, reducing unexpected breakdowns and optimizing maintenance efficiency. The results indicate that models like Gradient Boosting, Random Forest, and AdaBoost are highly effective in failure prediction, with the ROC curve analysis confirming their reliability.

For future work, the system can be enhanced by incorporating real-time streaming sensor data rather than relying solely on historical datasets. Additionally, integrating deep learning models such as LSTMs (Long Short-Term Memory networks) could further improve predictive accuracy by capturing complex temporal patterns. Expanding the dataset to include diverse aircraft engine types and environmental conditions would also enhance model generalization. Ultimately, the goal is to develop a fully automated, real-time predictive maintenance system that seamlessly integrates with aviation maintenance operations, ensuring higher safety and cost efficiency.

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- Ethics approval and consent to participate: This is an observational study. No ethical approval is required.
- Consent for publication: Not applicable.
- Author contribution: Davis Joseph, Sona Teresa Jossy, and Sonia Liz Jossy have contributed to the core research work. Manu Elappila has updated and reviewed the work, and guided the research.

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